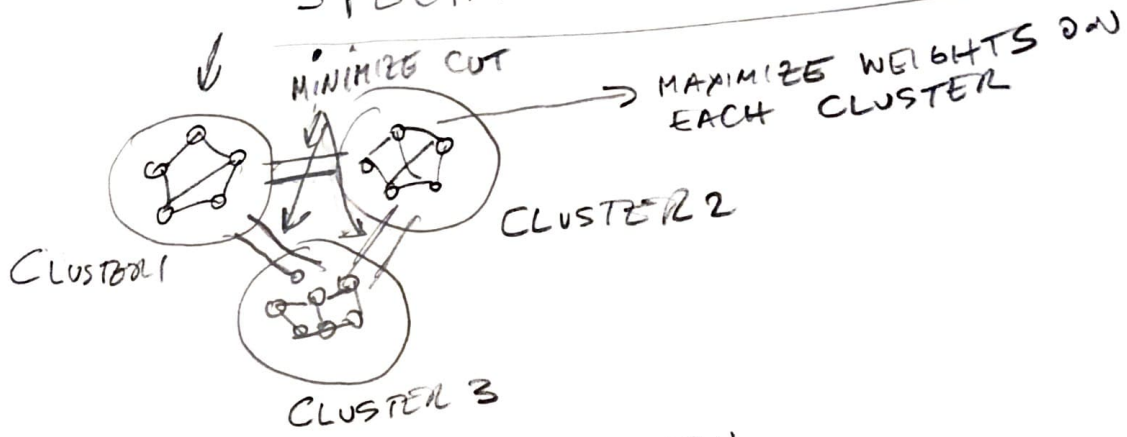
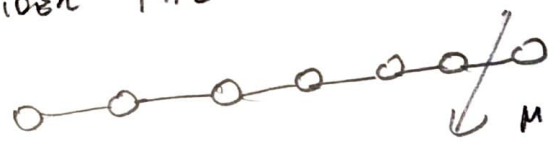
(1)

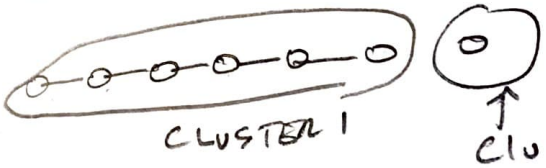
SPECTRAL CLUSTERING



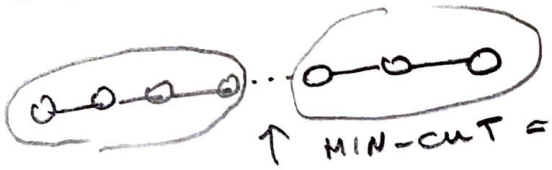
CONSIDER THE FOLLOWING GRAPH



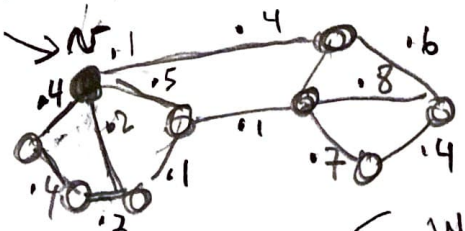
But we would like to balance the number of vertices on each cluster



We rather have a cut IN WHICH EACH CLUSTER HAVE EQUAL NUMBER OF VERTICES



ANOTHER EXAMPLE WITH WEIGHTS



The degree of v, $d_v = \sum w_{v,J}$, where $w_{v,J}$ is the weight of the edge (v, J).

Thus $d_v = 0.1 + 0.5 + 0.2 + 0.4 = 1.2$

G

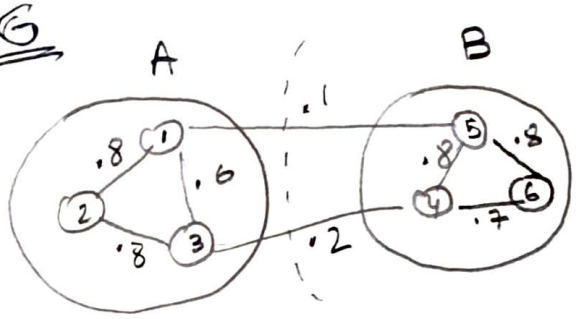


Fig 1

Cut (A, B) = sum of the weights CONNECTING A and B, thus Cut (A, B) = .3

Since we are talking about weights WE want To balance the SUM OF THE WEIGHTS ON EACH CLUSTER.

To find a minimum cut does not garantee balanced clusters.

Interesting enough we can use the laplacian Matrix To find a minimum cut while balancing the number of nodes of the clusters.

The Degree matrix for graph of Figure 1 is

$$D_G = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 1.5 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 1.6 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 1.6 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 & 1.7 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 & 1.7 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 & 1.5 \end{array}$$

The Adjacency matrix of the graph of Figure 1

$$W_G = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & .8 & .6 & 0 & .1 & 0 \\ 2 & .8 & 0 & .8 & 0 & 0 & 0 \\ 3 & .6 & .8 & 0 & .2 & 0 & 0 \\ 4 & 0 & 0 & .2 & 0 & .8 & .7 \\ 5 & .1 & 0 & 0 & .8 & 0 & .8 \\ 6 & 0 & 0 & 0 & .7 & .8 & 0 \end{array}$$

(3)

The laplacian $L_6 = D_6 - W_6$

$$L_6 = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 1.5 & -0.8 & -0.6 & 0 & -0.1 & 0 \\ 2 & -0.8 & 1.6 & -0.8 & 0 & 0 & 0 \\ 3 & -0.6 & -0.8 & 1.6 & -0.2 & 0 & 0 \\ 4 & 0 & 0 & -0.2 & 1.7 & -0.8 & -0.7 \\ 5 & -0.1 & 0 & 0 & -0.8 & 1.7 & -0.8 \\ 6 & 0 & 0 & 0 & -0.7 & -0.8 & 1.5 \end{array}$$

Some linear Algebra background

Let $\lambda_1, \lambda_2 \cdots \lambda_n$ be the eigenvalues of L_6

It can be proved that $\lambda_0 = 0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \lambda_n$

and the eigenvalues are non-negative real numbers.

The eigenvalue λ_i has an eigenvector x_i if

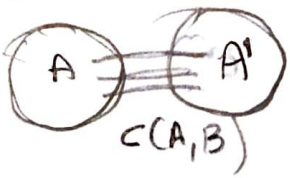
$$L_6 x_i = \lambda_i x_i$$

It can be proven that $\lambda_1 = 0$ and $x_1 = \mathbb{1}$

The second eigenvector, the Fiedler eigenvalue is called the "algebraic connectivity".

The Fiedler eigenvalue is related to a...

$$\lambda_2 \approx \min \frac{c(A, A')}{|A|}, |A| \leq |A'|$$



$|A|$ = # of vertices in A

4

Matrix calculator

Matrix calculator
Solving systems of linear equations
Determinant calculator
Eigenvalues calculator ✓
Wikipedia:Matrices

Hide Ads

Finding of eigenvalues and eigenvectors

This calculator allows to find eigenvalues and eigenvectors using the Characteristic polynomial .

Matrix A:

1.5	-0.8	-0.6	0	-0.1	0
-0.8	1.6	-0.8	0	0	0
-0.6	-0.8	1.6	-0.2	0	0
0	0	-0.2	1.7	-0.8	-0.7
-0.1	0	0	-0.8	1.7	-0.8
0	0	0	-0.7	-0.8	1.5

Cells | Clean | + | −

Find

More:

Diagonal matrix
Jordan decomposition
Matrix exponential

☐ Display decimals Clean

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Thanks to:

- Philip Petrov (https://cphpvb.net) for Bulgarian translation
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- Ousama Malouf and Yaseen Ibrahim for Arabic translation
- Marcel Artz - improving of the German translation
- Marc Gisbert Juàrez - fixing the translation into Catalan

Eigenvalues of L_6

$\lambda_1 = 0$
$\lambda_2 = 0.188$
$\lambda_3 = 2.084$
$\lambda_4 = 2.285$
$\lambda_5 = 2.469$
$\lambda_6 = 2.573$

Eigenvectors

$u_1 = \{1, 1, 1, 1, 1, 1\}$
$u_2 = \{-0.917, -0.993, -0.834, 0.834, 0.910, 1\}$
$u_3 = \{3.637, -0.081, -3.588, -1.904, 0.936, 1\}$
$u_4 = \{$
$u_5 = \{$
$u_6 = \{$

$L_6 u_2 = \lambda_2 \cdot u_2$

↓

Fiedler eigenvalue

Consider u_2 (the eigenvector of λ_2)

$u_2 = \{-0.917, -0.993, -0.834, 0.834, 0.910, 1\}$

Partition u_2 by finding $u_i \in u_2$ such that negative values are in one partition and non-negative values are in the other partition

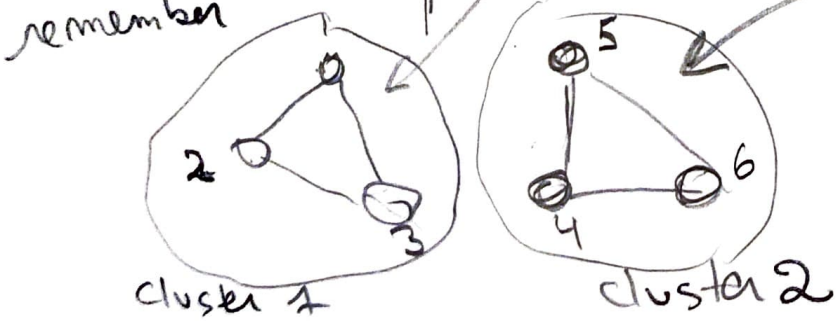
$\{-0.917, -0.993, -0.834\}$ $\{0.834, 0.910, 1\}$

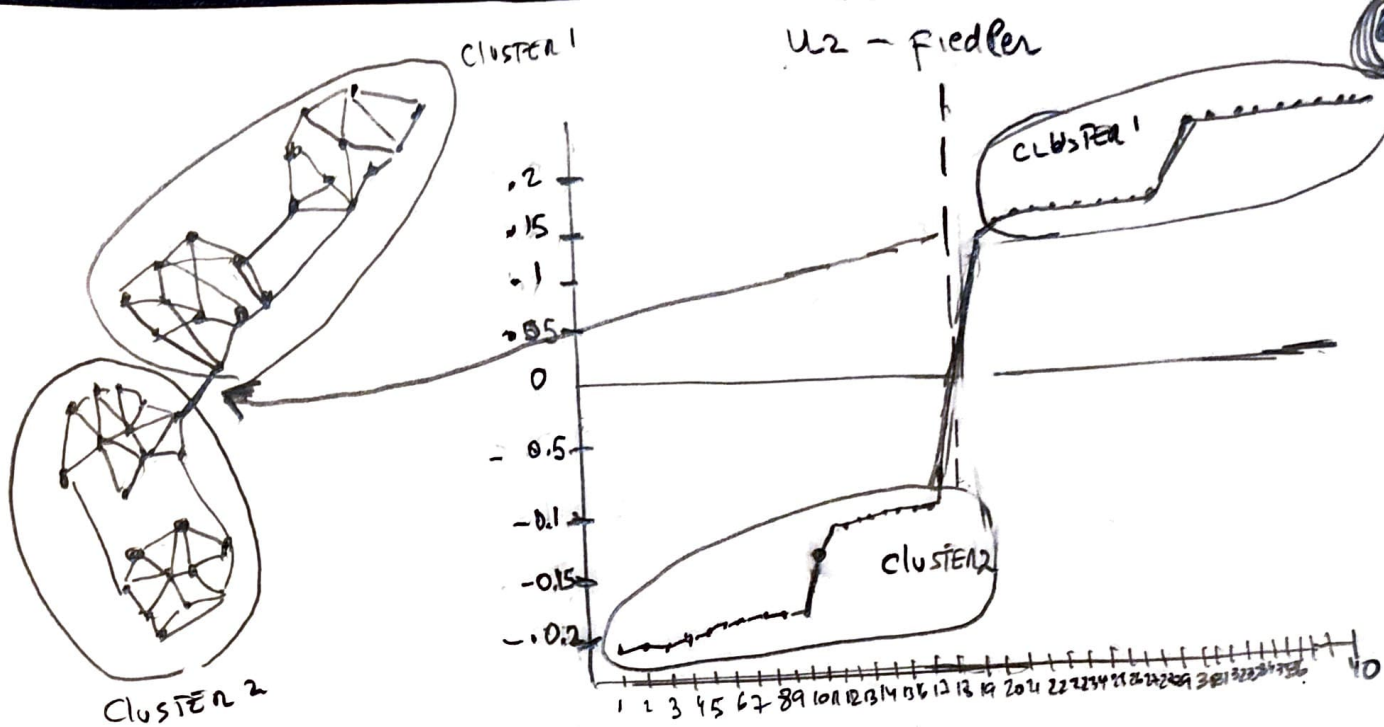
vertex 1 vertex 2 vertex 3 vertex 4 vertex 5 vertex 6

or plot



remember

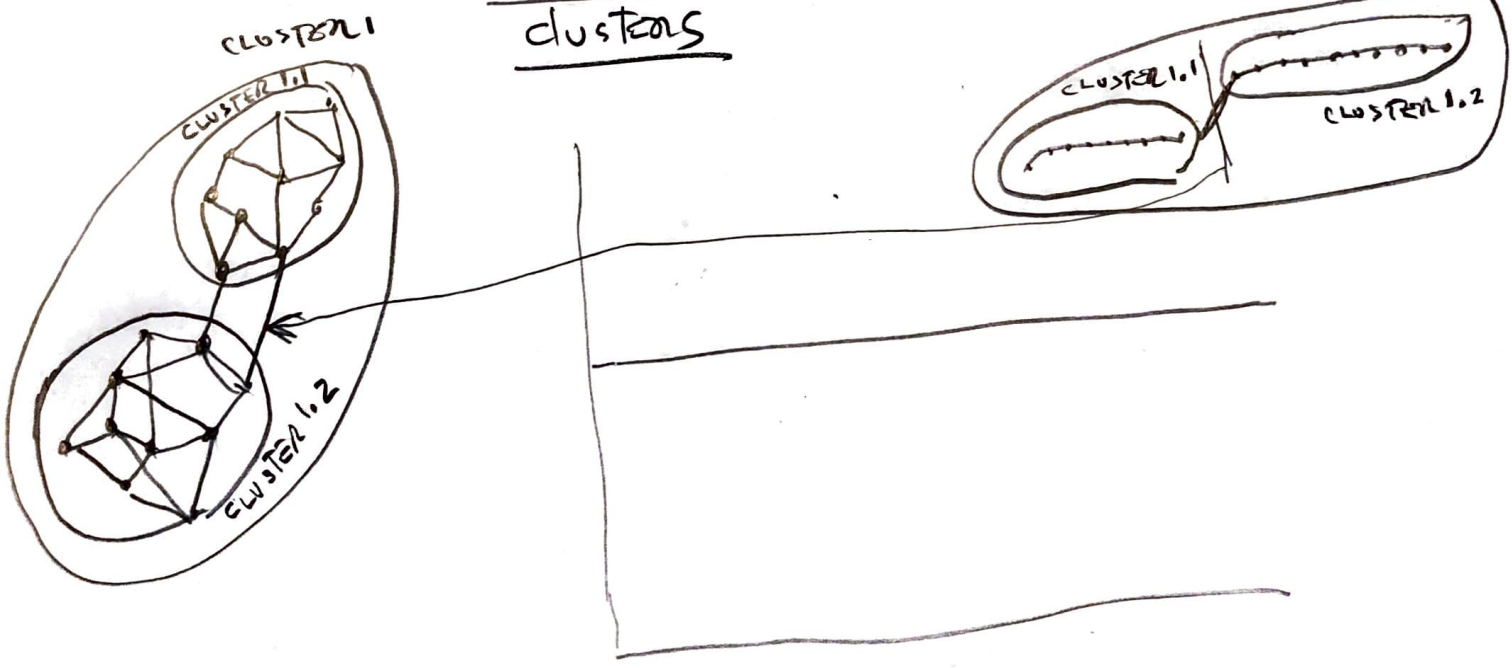




fiedler eigenvalue u2

Recursive bi-partition

Recursively break cluster 1 into two clusters



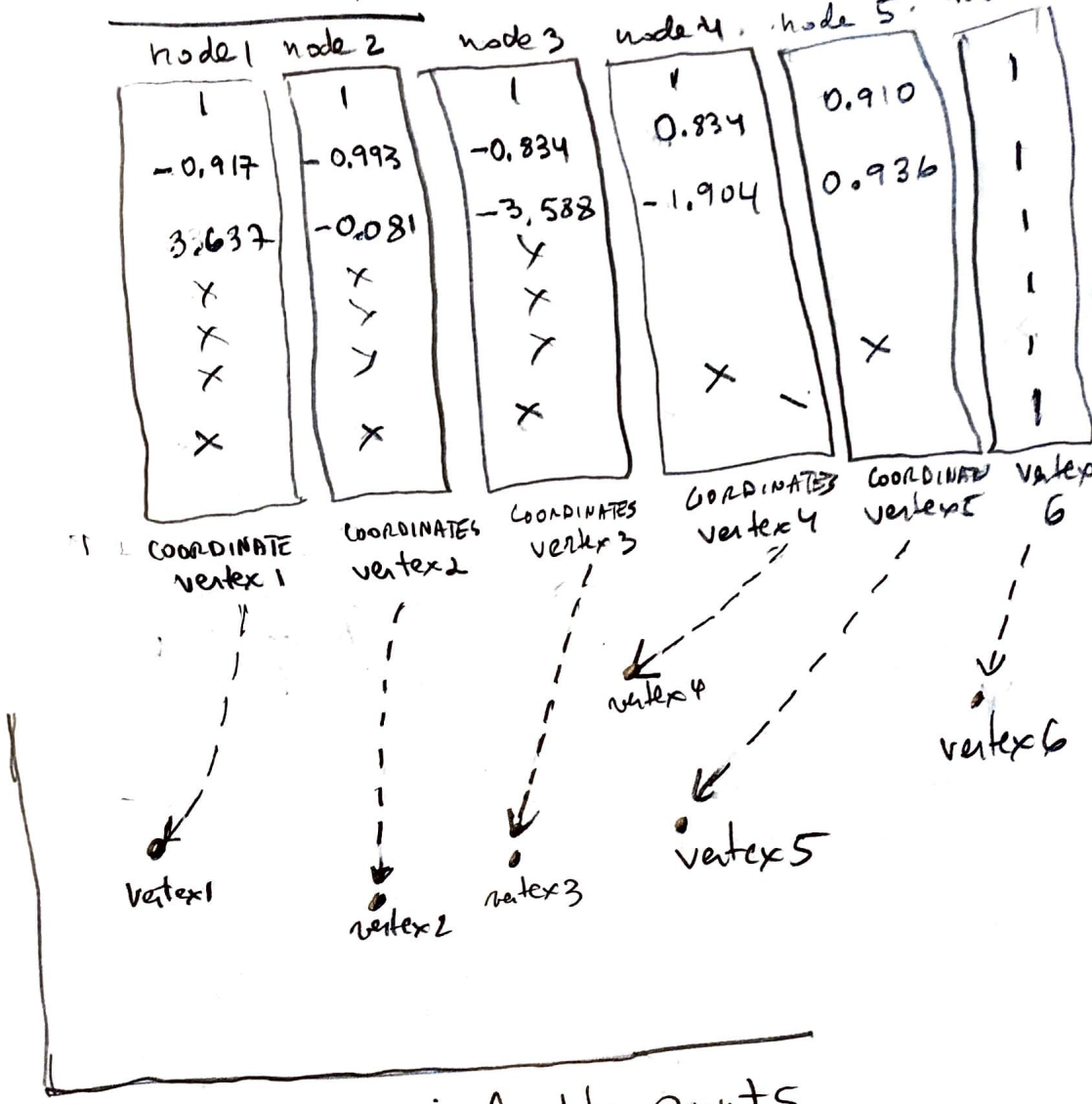
Another way to Find K-clusters

u_1 eigenvector

u_2 eigenvector

u_3 eigenvector

u_k eigenvector



Two Find the K clusters apply K-mean clustering